26900080 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

Regular EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSEBOTA06 - SOIL MICROBIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer type questions

Answer any **10** questions

Each question carries **2** marks

1. Mention the scope of soil microbiology. [U] / [CO1, CO2]
2. State the impact of no-tillage systems on soil microbial biomass. [U] / [CO5]
3. What is faecal coliform? [K] / [CO1]
4. Define agricultural waste. [K] / [CO5]
5. What are biopesticides? Name one fungal biopesticide. [K] / [CO5]
6. Name any two soil types favouring microbial growth. [K] / [CO1, CO2]
7. Define biofuel. Name one type of biofuel produced by microorganisms. [K] / [CO5]
8. What are airborne microorganisms? [K] / [CO1]
9. What is an infection thread? [K] / [CO1]
10. What is protozoan grazing? [K] / [CO1]

11. State the end products of microbial degradation of cellulose and lignin. [U] / [CO5]

12. Mention two important reservoirs of phosphorus. [K] / [CO1]

[2x10 = 20]

Part B

Short Essay type questions

Answer any **8** questions

Each question carries **5** marks

13. Explain the process of anaerobic sludge decomposition. Discuss the role of methanogens in methanogenesis. [U] / [CO5]

14. Describe the role of sulphur microorganisms in soil fertility. [K] / [CO1]

15. Describe the microbial breakdown of carbohydrates, proteins, lipids, lignin and cellulose. [U] / [CO3]

16. Describe the major factors that affect soil formation. [U] / [CO2, CO3]

17. List the environmental significance of nitrogen transformations. [K] / [CO1]

18. What are the key indicators of compost maturity? Why is it important to use mature compost in agriculture? [U] / [CO5]

19. Describe the types and applications of biofertilizers. Evaluate their advantages over chemical fertilizers. [U] / [CO5]

20. Describe the sources and distribution of microorganisms on leaf surfaces. [U] / [CO3]

21. How oxygen availability in soil influence microbial respiration. [U] / [CO1, CO2]

22. Explain the meaning of mineralization and soil organic carbon formation. [K] / [CO1]

[5x8 = 40]

Part C

Essay-type questions

Answer any **1** questions

Each question carries **10** marks

23. Define soil enzymes. Explain in detail about diffeerent soil enzymes and their functions. [U] / [CO2, CO3]

24. Explain the process of organic matter decomposition and describe mineralization and the formation of soil organic carbon. [K] / [CO1]

[10x1 = 10]